Student Academic Performance Dataset

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A Research Paper for Data Foundations in Machine Learning

Practical Assignment — Lessons 1–3 | Submitted: June 20, 2026

1. Where This Data Came From

Dataset Title: Student Academic Performance Dataset (Hargeisa, Somaliland — June 2026)

We collected this data over the last two days (18-19) in June 2026 by surveying students at two universities in Hargeisa, Somaliland. We wanted to understand what real-world factors might influence how well students do on their final exams.

How We Collected It

We handed out paper questionnaires to 55 undergraduate students, asking them about their demographics, study habits, and daily logistics — things like how far they live from campus and whether they have internet access. Then we pulled their final exam scores from official academic records with their permission.

The questionnaire had four sections:

- **Section A — Who they are:** age, gender, family income
- **Section B — How they study:** daily study hours, sleep hours, number of subjects
- **Section C — Daily logistics:** distance to school, internet access, part-time job status
- **Section D — Academic history:** previous semester GPA, attendance rate, final exam score

2. What's in the Dataset

There are 12 variables total: 11 features that might predict exam performance, and 1 target variable (the exam score itself).

Table

Feature	**Type**	**What It Is**	**Role**
age	Number	Student age (18–23 years)	Input
gender	Category	Male or Female	Input
study_hours_per_day	Number	Average daily study time (hours)	Input
attendance_rate	Number	Percentage of classes attended	Input
internet_access	Category	Has internet access? (Yes/No)	Input
part_time_job	Category	Works part-time? (Yes/No)	Input
distance_to_school_km	Number	Distance from home to campus (km)	Input
num_subjects	Number	Number of subjects enrolled	Input
sleep_hours	Number	Average nightly sleep (hours)	Input
family_income_category	Category	Low / Middle / High	Input
previous_gpa	Number	GPA from previous semester (0–4.0)	Input
final_exam_score	**Number**	**Final exam result (0–100)**	**Label (y)**

The target (y): `final_exam_score` is what we want our model to predict. It ranges from 46 to 95 in this dataset.

The predictors (X): The other 11 columns are our inputs. Some are continuous numbers (like study hours and GPA), some are discrete counts (like age and sleep hours), and some are categories (like gender and income level).

3. Dataset Overview

- **Size:** 55 students × 13 columns (1 ID + 11 features + 1 target)
- **Format:** CSV file
- **Missing data:** Just 1 cell — the study hours for student ID 15

  Here's a sample of 10 students (showing 7 of the 13 columns). Notice Row 15 has a blank for study hours:

Table

ID	Age	Gender	Study Hrs/Day	Attendance%	Prev GPA	Final Score
1	19	Male	3.5	85	2.8	74
2	20	Female	5.0	92	3.5	88
3	18	Male	2.0	70	2.2	61
4	21	Female	4.5	95	3.2	85
5	22	Male	1.5	60	1.9	52
6	19	Female	6.0	98	3.8	93
7	20	Male	3.0	78	2.5	67
15	18	Male	N/A	72	2.0	58
27	20	Male	1.0	58	1.6	46
40	23	Female	5.5	97	3.9	95

Table 1: Sample rows from the Student Academic Performance Dataset (7 of 13 columns shown)

4. Data Quality Issues

Real-world data is messy. Here's what we need to clean up before we can build any models:

4.1 Missing Values

Student 15 didn't fill in their study hours. This looks like a random omission (Missing Completely At Random), so we can safely fill it in using the average or median study time from the other students.

4.2 Text Categories

Three columns — `gender`, `internet_access`, and `part_time_job` — contain text like "Yes"/"No" and "Male"/"Female". Machine learning models need numbers, not words, so we'll need to convert these using label encoding or one-hot encoding.

4.3 Ordered Categories

Family income has a natural order: Low → Middle → High. Since there's a meaningful ranking here, ordinal encoding (Low=0, Middle=1, High=2) makes more sense than one-hot encoding.

4.4 Different Scales

Our numerical features are all over the place: study hours range from 0–6, age from 18–23, distance from 0.5–14.2 km, and attendance from 55–98%. Algorithms like K-Nearest Neighbors or SVMs will get confused by these wildly different scales, so we'll need to normalize or standardize everything.

4.5 No Duplicates

Good news: each student has a unique ID, and we didn't find any duplicate records.

4.6 Score Distribution

Final exam scores are spread across 46–95, but they tend to cluster in two groups: students with low study hours and poor attendance scoring 46–65, and high-attendance students scoring 74–95. This bimodal pattern is worth keeping in mind when we explore the data.

5. Why This is a Supervised Learning Problem

This dataset is perfect for **supervised learning** because we have a clear target: `final_exam_score`. Every student comes with both input features and a known outcome, giving us labeled training examples.

For supervised learning, we need:

- **Input features (X)** — check, we have 11 of them
- **Known output labels (y)** — check, we have exam scores for everyone
- **A goal of mapping X → y** — specifically, predicting a student's exam score

Since the target is a continuous number (0–100), this is a **regression** problem, not classification.

Unsupervised learning wouldn't fit our main goal. We could use clustering as a side analysis to find natural student groups, but if we want to predict scores, we need supervision.

6. How We'd Use This in a Real ML Project

6.1 The Machine Learning Task

The main task is **regression**: given a student's profile, predict their final exam score. Some algorithms to try:

- **Linear Regression** — simple and interpretable

- **Decision Tree Regressor** — catches non-linear patterns
- **Random Forest Regressor** — robust ensemble approach
- **Support Vector Regressor** — works well with normalized data

We could also turn this into a **classification** problem by grouping scores into categories (e.g., Fail < 60, Pass 60–79, Distinction ≥ 80), then try Logistic Regression, K-Nearest Neighbors, or a Decision Tree Classifier.

6.2 Where This Fits in the Data Science Lifecycle

Table

Stage	What We Do	How This Dataset Applies
1. Problem Definition	Identify what we want to know	What factors affect student exam performance?
2. Data Collection	Gather raw data	Done — 55 students surveyed, exam records gathered
3. Data Preprocessing	Clean and prepare data	Handle missing value, encode categories, scale features
4. Exploratory Data Analysis	Explore and visualize	Check correlations (e.g., study hours vs. score), spot outliers
5. Modeling	Train a model	Train regression model; evaluate with RMSE and R^2
6. Deployment / Insight	Share findings	Give recommendations to educators on which factors predict success

6.3 Real-World Impact

A model trained on this data could help universities spot at-risk students early — maybe someone who lives far from campus, studies little, or lacks internet access — and get them support before finals.

End of Research Paper — Dataset file: `student_study_dataset.csv`